



Advanced Log Analysis, Threat Intelligence Integration, Incident Escalation, Evidence Preservation, and SOC Workflow Simulation

1. Introduction

Security Operations Centers (SOCs) play a critical role in monitoring, detecting, analyzing, and responding to cybersecurity incidents. Modern organizations generate large volumes of security logs from endpoints, servers, firewalls, applications, and cloud services. Security analysts must continuously review these logs, identify suspicious behavior, investigate threats, and coordinate response activities.

This assignment focuses on practical SOC operations including advanced log analysis, threat intelligence integration, incident escalation procedures, alert triage, evidence preservation, and complete incident response workflows. The activities performed during this assignment simulate real-world SOC responsibilities and provide hands-on exposure to commonly used security tools and methodologies.

The primary goal of this exercise is to understand how security events are detected, analyzed, enriched with intelligence, escalated appropriately, and documented throughout the incident lifecycle.

2. Objectives

The objectives of this assignment are:

- Understand advanced log analysis techniques.
- Learn log correlation and anomaly detection.
- Integrate threat intelligence feeds into security monitoring activities.
- Practice threat hunting using Indicators of Compromise (IOCs).
- Understand SOC escalation workflows and communication procedures.
- Perform alert triage and IOC validation.
- Learn evidence collection and preservation techniques.
- Simulate a complete security incident from detection to reporting.
- Improve analytical, investigative, and reporting skills required in a Security Operations Center environment.

3. Tools and Technologies Used

The following tools were utilized during this assignment:



Elastic Stack (ELK)

Elastic Stack consists of Elasticsearch, Logstash, Kibana, and Beats. It provides centralized log collection, storage, visualization, and analysis capabilities.

Wazuh

Wazuh is an open-source security monitoring platform used for log analysis, intrusion detection, compliance monitoring, and threat detection.

AlienVault OTX

AlienVault Open Threat Exchange (OTX) provides threat intelligence feeds containing malicious IP addresses, domains, hashes, and other indicators.

TheHive

TheHive is an incident response platform used for case management, investigation tracking, and collaboration.

VirusTotal

VirusTotal is a threat intelligence platform used to validate files, IP addresses, domains, and URLs against multiple security engines.

Velociraptor

Velociraptor is a digital forensic and incident response tool used to collect system artifacts and perform investigations.

Metasploit Framework

Metasploit is a penetration testing framework used to simulate attacks and validate security monitoring capabilities.

CrowdSec

CrowdSec is an open-source security engine used for detecting and blocking malicious activities.

4. Advanced Log Analysis

Overview

Advanced log analysis involves collecting, reviewing, and correlating logs from multiple sources to identify security incidents. Security analysts rely on log analysis to detect malicious behavior, investigate incidents, and understand attacker activity.

Log Correlation

Log correlation is the process of linking multiple events from different sources to identify meaningful patterns. Individual events may appear harmless when viewed separately; however, when combined, they can reveal suspicious activity.

Example:

A failed login attempt followed by outbound network communication may indicate credential abuse or unauthorized access attempts.

Sample Correlation Scenario

Timestamp	Event ID	Source IP	Destination IP	Description
12:00	4625	192.168.1.100	N/A	Failed Login
12:01	4625	192.168.1.100	N/A	Failed Login
12:02	5156	192.168.1.100	104.21.12.50	Outbound Connection

The sequence suggests potential brute-force activity followed by suspicious communication.

Anomaly Detection

Anomaly detection identifies unusual behavior that deviates from normal activity.

Examples include:

- Large data transfers
- Excessive failed logins
- Access during non-business hours
- Unusual process execution

Anomaly detection helps identify emerging threats that may not match known attack signatures.

Log Enrichment

Log enrichment enhances raw log data by adding contextual information such as:

- Geographic location
- Asset ownership
- Threat intelligence information
- User roles

GeoIP enrichment enables analysts to identify the geographic origin of network traffic and improve investigation efficiency.

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5. Threat Intelligence Integration

Overview

Threat intelligence provides information about known malicious indicators and attacker behavior. Integrating threat intelligence into a SOC improves detection accuracy and accelerates investigations.

Indicators of Compromise (IOCs)

Indicators of Compromise are artifacts that suggest malicious activity.

Examples include:

- Malicious IP addresses
- Domains
- File hashes
- URLs

Threat Intelligence Feeds

Threat intelligence feeds continuously provide updated information regarding emerging threats.

Sources include:

- AlienVault OTX
- VirusTotal
- Commercial Intelligence Providers

MITRE ATT&CK Framework

MITRE ATT&CK is a globally recognized framework that documents adversary tactics and techniques.

Example:

T1078 – Valid Accounts

Attackers use legitimate credentials to gain unauthorized access.

Threat Hunting

Threat hunting involves proactively searching for indicators of malicious activity before alerts are generated.

Benefits include:

- Early threat detection



- Reduced dwell time
- Improved security posture

Findings

Threat intelligence integration significantly improves detection accuracy and provides additional context for security investigations.

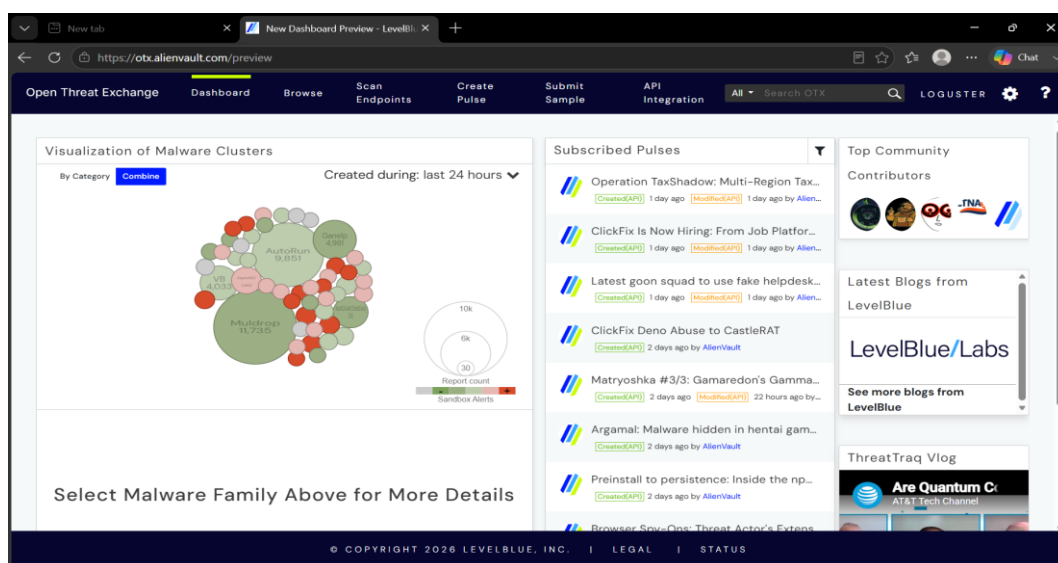


Figure 3: AlienVault OTX threat intelligence dashboard displaying community threat feeds and malware intelligence sources used for IOC enrichment.

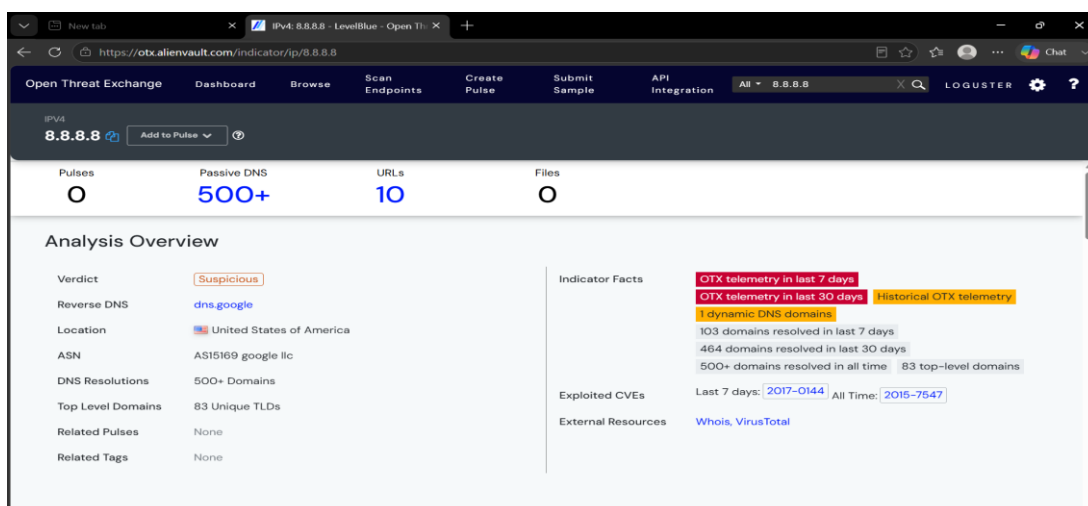


Figure 4: Threat intelligence analysis of IP address 8.8.8.8 using AlienVault OTX. The lookup provides contextual information, DNS records, and reputation details.



6. Incident Escalation Practice

Overview

Incident escalation ensures that security incidents are investigated by personnel with appropriate expertise.

Tier 1 Responsibilities

- Alert monitoring
- Initial triage
- Basic investigation

Tier 2 Responsibilities

- Deep investigation
- Containment activities
- Root cause analysis

Tier 3 Responsibilities

- Advanced threat analysis
- Malware reverse engineering
- Threat hunting

Escalation Workflow

1. Alert Generated
2. Initial Triage
3. Severity Assessment
4. Evidence Collection
5. Escalation to Tier 2
6. Investigation
7. Remediation
8. Closure

Situation Reporting (SITREP)

Effective communication is essential during incident response.

A SITREP should include:



- Incident Summary
- Affected Systems
- Current Status
- Actions Taken
- Recommended Next Steps

Findings

Proper escalation workflows reduce response times and ensure incidents receive appropriate attention.

Incident Title: Unauthorized Access on Server-Y
Severity: High
Detected Time: 2025-08-18 13:00
MITRE ATT&CK: T1078
Action Taken:
- Investigated by Tier 1
- Escalated to Tier 2
- Evidence Collected
Status: Under Investigation

Figure 5: Situation Report (SITREP) prepared for an unauthorized access incident. The report summarizes affected systems, actions taken, current status, and recommended next steps.

SITUATION REPORT (SITREP)	
Incident: Unauthorized Access on Server-Y	
Summary: Suspicious authentication activity detected from IP address 192.168.1.200.	
Affected System: Server-Y	
MITRE ATT&CK: T1078 – Valid Accounts	
Actions Taken: - Server isolated - Logs collected - Escalated to Tier 2	
Current Status: Investigation in Progress	
Next Steps: - Review authentication logs - Validate source IP - Assess impact	

Figure 6: Incident escalation documentation showing incident details, severity classification, investigation status, and escalation to Tier 2 analysts.



7.Alert Triage with Threat Intelligence

Overview

Alert triage involves determining whether an alert represents legitimate malicious activity.

Alert Validation Process

1. Review Alert Details
2. Identify Indicators
3. Validate Indicators
4. Assess Risk
5. Assign Priority
6. Escalate if Necessary

IOC Validation

Threat intelligence sources such as VirusTotal and AlienVault OTX help determine whether indicators are malicious.

Risk Prioritization

Alerts are categorized based on:

- Severity
- Confidence
- Business Impact
- Exploitability

Findings

Threat intelligence validation improves analyst confidence and reduces false positives.

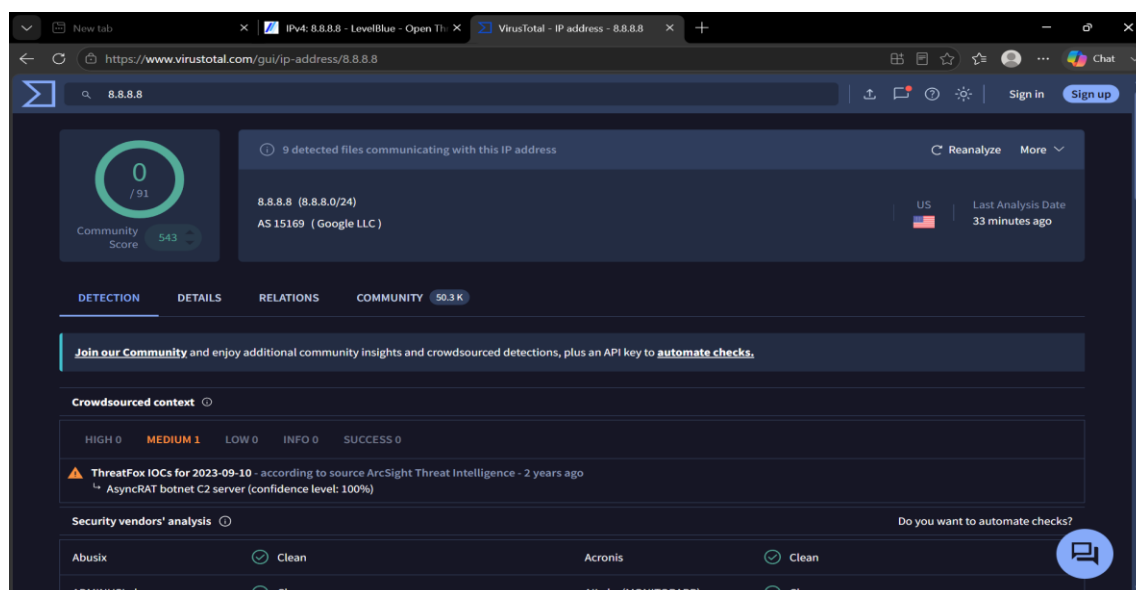


Figure 7: VirusTotal reputation analysis used to validate indicators of compromise and verify the security status of an investigated IP address.

8.Evidence Preservation and Analysis

Overview

Evidence preservation is critical for incident investigation and legal compliance.

Volatile Data Collection

Volatile data includes:

- Running processes
- Network connections
- Memory contents

This information should be collected before system shutdown.

Memory Acquisition

Memory analysis can reveal:

- Malware
- Active sessions
- Encryption keys
- Network connections



Hashing and Integrity Verification

SHA256 hashing ensures evidence remains unchanged after collection.

Benefits include:

- Evidence integrity
- Chain of custody validation
- Forensic reliability

Findings

Evidence preservation supports accurate investigations and ensures forensic reliability.

```
kali@kali: ~/Desktop
File Actions Edit View Help
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53526 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53416 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53802 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53262 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53518 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53264 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53740 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:34058 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53724 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53232 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53374 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:34056 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53680 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53400 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53728 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53818 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53804 ESTABLISHED
tcp6 0 0 127.0.0.1:9200 127.0.0.1:53674 ESTABLISHED

(kali@kali)~$ echo "SOC Evidence Collection" > evidence.txt
(kali@kali)~$ cat evidence.txt
SOC Evidence Collection
(kali@kali)~$ sha256sum evidence.txt
7421b21f663567f4de1b81c44a24fee581d7a9983fd19abc99d43b2190589110 evidence.txt
(kali@kali)~$ sha256sum evidence.txt
7421b21f663567f4de1b81c44a24fee581d7a9983fd19abc99d43b2190589110 evidence.txt
(kali@kali)~$
```

Figure 8: Collection of volatile forensic evidence from the investigation environment. Evidence was preserved to support incident analysis and maintain integrity.

9. Full SOC Workflow Simulation

Overview

The capstone exercise simulates the complete incident lifecycle from detection to reporting.

Attack Simulation

A controlled attack was executed against a vulnerable target system to validate monitoring and response capabilities.



Detection

Security monitoring tools generated alerts indicating suspicious activity.

Investigation

Analysts reviewed logs, validated indicators, and assessed the severity of the incident.

Containment

Response actions included:

- Host isolation
- IP blocking
- Threat eradication

Escalation

The incident was escalated according to SOC procedures for advanced investigation.

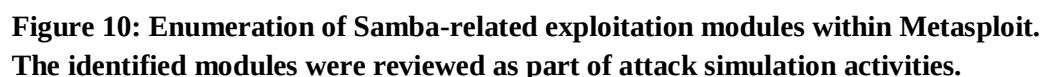
Reporting

An incident report was prepared documenting:

- Executive Summary
- Timeline
- Findings
- Recommendations

Findings

The simulation demonstrated the effectiveness of SOC monitoring, investigation, containment, and reporting processes.



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- Alert triage reduces false positives.
- Evidence preservation supports forensic investigations.
- End-to-end simulations strengthen SOC readiness.

11. Challenges Faced

During the assignment several challenges were encountered:

- Initial ELK Stack configuration issues.
- Filebeat authentication errors.
- Data ingestion troubleshooting.
- Threat intelligence feed integration.
- Understanding incident escalation workflows.

These challenges provided valuable learning opportunities and improved troubleshooting skills.

12. Conclusion

This assignment provided practical exposure to Security Operations Center operations and incident response methodologies. Through log analysis, threat intelligence integration, alert triage, incident escalation, evidence preservation, and workflow simulation, a deeper understanding of SOC processes was achieved. The knowledge gained from these activities strengthens the foundation required for professional cybersecurity and SOC analyst roles.

13. References

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